

Perdiction Results

2024-10-02

Data Summary:

- **Number of observations for bill passage predictions:** 40,000. No null responses (using v3 of prompts)
- **Number of observations for bill summary completion:** 39,995. A total of 45 responses exceeded the token limit of 750, and 5 null responses were dropped.

Results

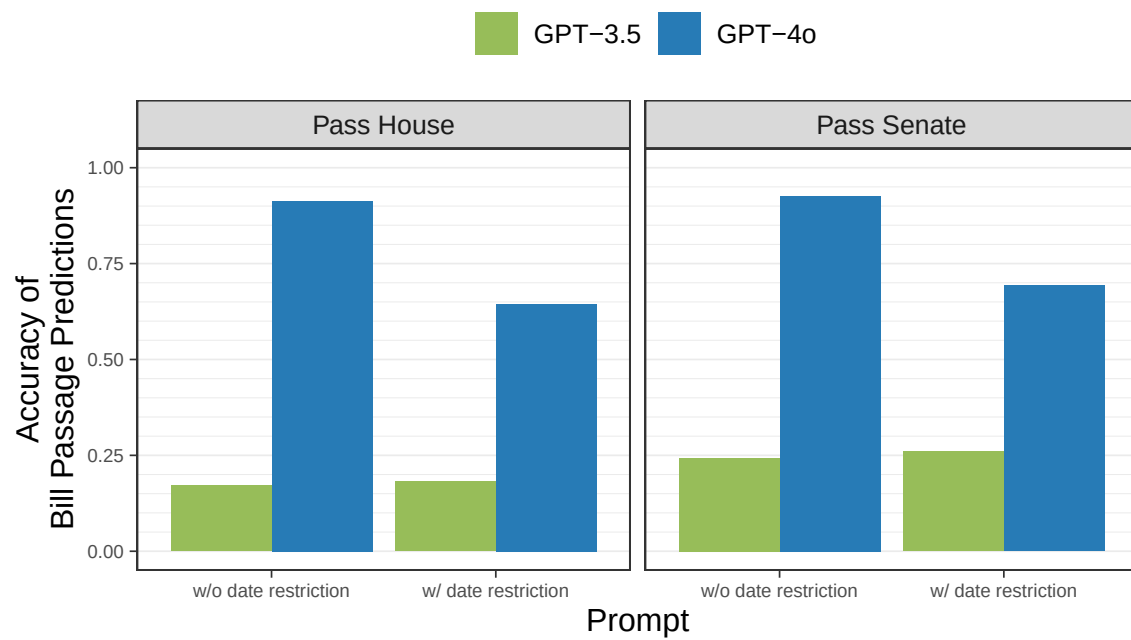
Bill Passage

Table 1: True Bill Passage Rates

Chamber	True Passage Rate
Pass House	0.074
Pass Senate	0.058

Table 2: Performance Evaluation of LLM Predictions of Bill Passage

Predicted Outcome	Model	Prompt	Accuracy	TPR	FPR
Pass House	GPT-3.5	w/o date restriction	0.171	0.982	0.894
Pass House	GPT-3.5	w/ date restriction	0.182	0.977	0.882
Pass House	GPT-4o	w/o date restriction	0.913	0.190	0.029
Pass House	GPT-4o	w/ date restriction	0.644	0.715	0.361
Pass Senate	GPT-3.5	w/o date restriction	0.243	0.965	0.801
Pass Senate	GPT-3.5	w/ date restriction	0.261	0.967	0.782
Pass Senate	GPT-4o	w/o date restriction	0.927	0.212	0.030
Pass Senate	GPT-4o	w/ date restriction	0.693	0.743	0.310



Bill Summary Completion

Table 3: Performance Evaluation of LLM Completion of Bill Summaries (Non-Embedded)

Model	Prompt	Average
GPT-3.5	w/o date restriction	0.014
GPT-3.5	w/ date restriction	0.015
GPT-4o	w/o date restriction	0.043
GPT-4o	w/ date restriction	0.038

Table 4: Performance Evaluation of LLM Completion of Bill Summaries (Embedded)

Metric	Model	Prompt	Average	Baseline	Relative to Baseline
Cosine Similarity	GPT-3.5	w/o date restriction	0.489	0.244	1.009
Cosine Similarity	GPT-3.5	w/ date restriction	0.499	0.245	1.037
Cosine Similarity	GPT-4o	w/o date restriction	0.538	0.244	1.204
Cosine Similarity	GPT-4o	w/ date restriction	0.542	0.244	1.226
Euclidean Distance	GPT-3.5	w/o date restriction	0.986	1.228	0.197
Euclidean Distance	GPT-3.5	w/ date restriction	0.976	1.227	0.205
Euclidean Distance	GPT-4o	w/o date restriction	0.918	1.228	0.252
Euclidean Distance	GPT-4o	w/ date restriction	0.917	1.228	0.254

